

Entraining Power-Dropout Events in an External-Cavity Semiconductor Laser Using Weak Modulation of the Injection Current

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Abstract—We measure experimentally the effects of injection current modulation on the statistical distribution of time intervals between power-dropout events occurring in an external-cavity semiconductor laser operating in the low-frequency fluctuation regime. These statistical distributions are sensitive indicators of the presence of pump current modulation. Under most circumstances, we find that weak low-frequency (in the vicinity of 19 MHz) modulation of the current causes the dropouts to occur preferentially at intervals that are integral multiples of the modulation period. The dropout events can be entrained by the periodic perturbations when the modulation amplitude is large (peak-to-peak amplitude $\geq 8\%$ of the dc injection current). We conjecture that modulation induces a dropout when the modulation frequency is equal to the difference in frequency between a mode of the extended cavity laser and its adjacent antinode. We also find that the statistical distribution of the dropout events is unaffected by the periodic perturbations when the modulation frequency is equal to the free spectral range of the external cavity. Numerical simulations of the extended-cavity laser display qualitatively similar behavior. The relationship of these phenomena to stochastic resonance is discussed and a possible use of the modulated laser dynamics for chaos communication is described.

Index Terms—Chaos communication, controlling chaos, entrainment, external-cavity semiconductor lasers, low-frequency fluctuations, power-dropout events.

I. INTRODUCTION

SEMICONDUCTOR lasers in the presence of delayed optical feedback exhibit a wide variety of dynamical behaviors arising from a complex interaction of deterministic and stochastic forces. Such external-cavity semiconductor laser (ECSL) systems have attracted a great deal of attention because this configuration appears frequently in communication and data storage technologies (where unwanted reflections typically arise from a fiber facet or optical disk), and because of an interest in understanding fundamental semiconductor laser dynamics and time-delay systems. Under a variety of conditions, the external optical feedback induces instabilities and chaos in the laser dynamics, thereby increasing the amplitude

fluctuations and reducing the coherence of the light generated by the laser. This behavior results in reduced device performance for many applications [1], [2]; hence, it is important to devise methods for avoiding or dynamically controlling the instabilities.

One particular unstable behavior is known as “low-frequency fluctuations” (LFF’s) [3], [4], often occurring when the laser operates near threshold and is subjected to moderate optical feedback from an external cavity in which the round-trip time is much longer than the period of the solitary laser’s relaxation oscillation frequency. In the LFF state, the laser produces an erratic train of ultrashort pulses [5]–[7], each approximately 100 ps wide, with spacings ranging between 200–1000 ps. This behavior is interrupted sporadically at much longer intervals by power-dropout events, during which the average power of the light generated by the laser suddenly drops, then gradually builds up to its original value over approximately ten external-cavity round-trip times. The power-dropout events result in greatly increased noise in the radio frequency spectrum, as observed by Risch and Voumard [3] over two decades ago; ongoing research has focused on understanding the mechanisms and influences responsible for LFF’s (see, for example, [5]–[20]).

Recent research from the nonlinear dynamics community suggests that it may be possible to control the LFF instability by applying only small perturbations to an accessible system parameter, such as the injection current, for example. The key idea underlying this new class of control schemes is to take advantage of the unstable steady states and unstable periodic orbits of the system. The control protocols attempt to stabilize one such unstable state by making small adjustments to an accessible system parameter when the system is in a neighborhood of the state.

Techniques for stabilizing unstable states in nonlinear dynamical systems using small perturbations fall into two general categories: feedback and nonfeedback schemes. The idea that chaos and instabilities can be controlled efficiently using feedback (closed-loop) schemes to stabilize unstable states was suggested by Ott, Grebogi, and Yorke [21] in 1990, and many adaptations of their original concept have been investigated theoretically [22]–[30] and experimentally [31]–[36] for controlling laser dynamics. One serious issue that arises when attempting feedback control of the LFF instability is the fast time scale of the dynamics. Sukow *et al.* [37] have shown that the ability to control a fast instability is lost or severely degraded when the control-loop latency (the time between measuring the dynamical state of the laser and the application of the perturbation) is

Manuscript received February 24, 1998; revised September 8, 1999. This work was supported by the National Research Council and the Air Force Office of Scientific Research. The work of D. J. Gauthier was supported by the Army Research Office under Grant DAAD19-99-1-0199 and Grant DAAG55-97-1-0308.

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Publisher Item Identifier S 0018-9197(00)00958-1.

larger than or comparable to the time characteristic of the instability [38]. In the case of LFF dynamics, this characteristic time may be as brief as tens of picoseconds.

The problems associated with control-loop latency can be avoided by using nonfeedback (open-loop) schemes, where an orbit similar to the desired unstable state is entrained by adjusting an accessible system parameter about its nominal value by a weak periodic signal, usually in the form of a continuous sinusoidal modulation. This is somewhat simpler than feedback schemes because it does not require real-time measurement of the state of the system and processing of a feedback signal. Unfortunately, periodic modulation fails in many cases to entrain the unstable state; its success or failure is highly dependent on the specific form of the dynamical system [39], and it is known that periodic modulation can *induce* LFF in a semiconductor laser [40]. Regardless, open-loop perturbations have been used successfully to entrain the dynamics of low-dimensional laser systems [41]–[43], usually by modulating a system parameter at a frequency that is present in the free-running system such as the relaxation oscillation frequency, for example. Nonfeedback perturbations have also been shown to entrain periodic behavior in some semiconductor laser systems [44]–[47]. Furthermore, from the standpoint of optical engineering, the high-frequency injection technique is a well-known tool for reducing the relative intensity noise in ECSL's [48], [49]. Thus, there is ample precedent to suspect that open-loop perturbations may also be able to restrict LFF dynamics in a useful way. Open-loop perturbation of LFF dynamics is also interesting from the standpoint of stochastic resonance, a phenomenon in which noise in a nonlinear system may enhance a weak periodic signal [50], [51]. Spontaneous emission noise is quite strong in semiconductor lasers and, thus, has the potential to affect the system's response to a periodic perturbation in unexpectedly strong ways.

The main purpose of this paper is to describe our experimental investigation of an ECSL operating in the LFF regime and subjected to a weak high-frequency periodic modulation of the injection current. The effect of the modulation on the laser dynamics is determined by measuring the probability distribution of the time τ between consecutive power-dropout events. We find that the distribution is extremely sensitive to weak perturbations. We first describe the apparatus and data acquisition techniques used in this experiment, followed by experimental results for two different modulation frequency ranges, and finally discuss our findings. The possible application of ECSL's with weak periodic modulation to emerging chaos communication schemes is mentioned briefly.

II. EXPERIMENTAL PROCEDURES

Experiments on LFF are complicated by the extraordinary range of time scales present, with interesting dynamics occurring over several orders of magnitude in time. A complete time series analysis would require data records several microseconds long and picosecond time resolution; these demands exceed the capabilities of state-of-the-art measurement equipment. Therefore, we perform instead a statistical study of the time between power-dropout events and characterize how this time is modified by weak periodic perturbations.

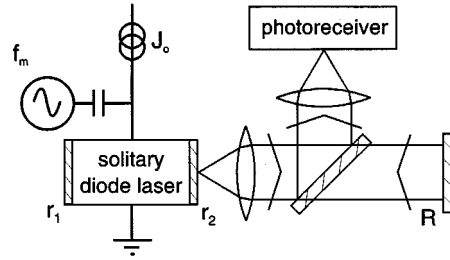


Fig. 1. Experimental setup.

A schematic diagram of the experimental apparatus is shown in Fig. 1. The laser diode (Spectra Diode model SDL-5401-G1) operates in a single longitudinal and transverse mode with a nominal wavelength of $\lambda = 789$ nm and threshold current of $I_{th} = 17$ mA in the absence of external feedback. It is mechanically isolated and temperature-stabilized to approximately ± 1 mK. The laser is pumped electrically through a bias-tee (Mini-Circuits model ZFBT-6GW), combining a constant dc current with a sinusoidal RF component from a leveled function generator (Tektronix model SG503). A high-numerical-aperture lens ($NA = 0.5$) collimates the laser's output field, which propagates through the external cavity of length $L = 71$ cm and is directed back to the laser by a high-reflectivity mirror. We position the external mirror for optimum alignment, minimizing the laser's current threshold in the presence of optical feedback. We place a polarizing beamsplitter cube and a rotatable quarter-wave plate in the external cavity to provide smooth adjustment of the optical feedback level. A plate beamsplitter in the beam path directs 30% of the laser output to a 6-GHz photoreceiver (New Focus model 1537-LF) that is connected to an RF spectrum analyzer (Tektronix model 2711) and a 1-GHz analog bandwidth digitizing oscilloscope (Tektronix model TDS680B). We note that the 1-GHz analog bandwidth of the oscilloscope smooths out most of the fast pulsing dynamics, resulting in a measurement of only the slower dropout envelope as desired for this study.

We acquire statistical data of the dropout intervals τ through an automated LabVIEW program. The program downloads a 5000-point time series from the oscilloscope over a GPIB interface, uses a peak-finding algorithm to extract sequential dropout intervals, and repeats this process until a sufficient number of intervals have been obtained (typically 10 000 events). For the purposes of this procedure, τ is defined as the time between the sharp falling edge at the beginning of a dropout and the same point on the next dropout. We measure τ with a resolution of approximately 1 ns in our experiment. After collecting all data points for a given parameter set, we calculate a probability density $\eta(\tau)$, where $\eta(\tau)\Delta\tau$ represents approximately the probability that a measured interval will fall between τ and $\tau + \Delta\tau$ in the limit as $\Delta\tau$ becomes small. The density $\eta(\tau)$ is determined by constructing a histogram of the dropout interval occurrences, scaled to the total number of points and the bin width of the histogram.

III. EXPERIMENTAL OBSERVATIONS: LOW-FREQUENCY MODULATION

In the first series of experiments, we investigate the effects on $\eta(\tau)$ due to a low-frequency RF modulation of the injection

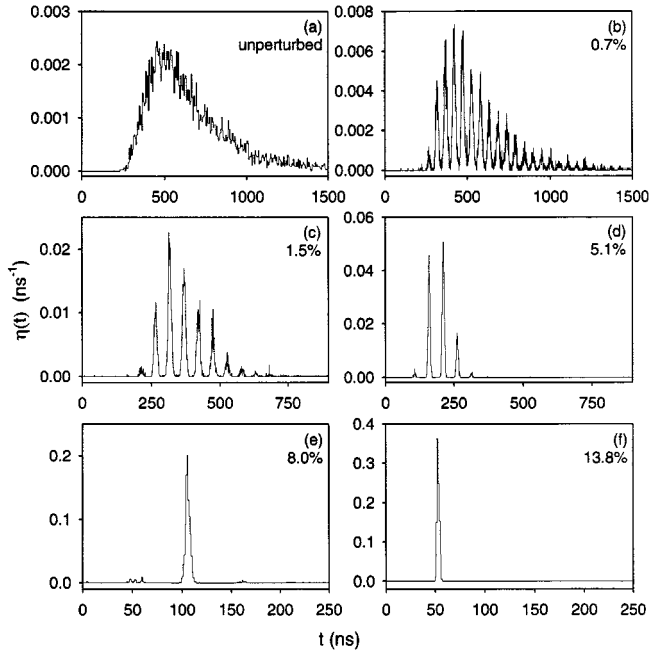


Fig. 2. Experimental probability distributions $\eta(\tau)$ of dropout intervals when a 19-MHz modulation is applied to the injection current. (a) Unperturbed distribution. Peak-to-peak modulation amplitudes for the others, expressed as a percentage of the dc level, are (b) 0.7%, (c) 1.5%, (d) 5.1%, (e) 8.0%, and (f) 13.8%.

current as the strength of the modulation is varied when the optical feedback is adjusted to reduce the laser threshold to 14.1 mA (a 17% reduction from its solitary value). Fig. 2 shows probability densities $\eta(\tau)$ as a function of increasing modulation amplitude at a frequency $f_m = 19$ MHz for a pump current level relative to threshold equal to $I/I_{th} = 1.29$. We choose 19 MHz because it produces entrainment of dropouts at the lowest drive amplitude (see below), although similar results are obtained for f_m in the range from 10 to 25 MHz. In the next section, we will motivate why such low-frequency modulation can entrain successfully the dropout events.

For reference, Fig. 2(a) shows the dropout interval statistics of the unperturbed system. The distribution shows a “dead zone” at short times (≤ 250 ns), then a rapid rise followed by a slowly decaying tail [17]. The average dropout interval $\langle\tau\rangle$ is 719 ± 9 ns, where the error in the measurement is assigned by assuming only statistical errors. This is a typical distribution shape, although other more structured forms have been observed for other operating conditions [17].

Fig. 2(b) shows the ECSL’s response to a very weak modulation, where the peak-to-peak modulation amplitude is only 0.7% of the dc level. The change in $\eta(\tau)$ is striking. The previously smooth distribution now has a comb-like structure, where the peaks are separated in time by multiples of the modulation period ~ 53 ns. This indicates that the power dropouts now occur preferentially at the same phase in the drive cycle, even though the average dropout interval is about ten drive cycles long. Interestingly, the overall envelope of $\eta(\tau)$ is not changed appreciably and the average interval only decreases to $\langle\tau\rangle = 559 \pm 6$ ns. This distribution is qualitatively similar to those obtained numerically by Mulet [52] using a modulated Lang–Kobayashi model.

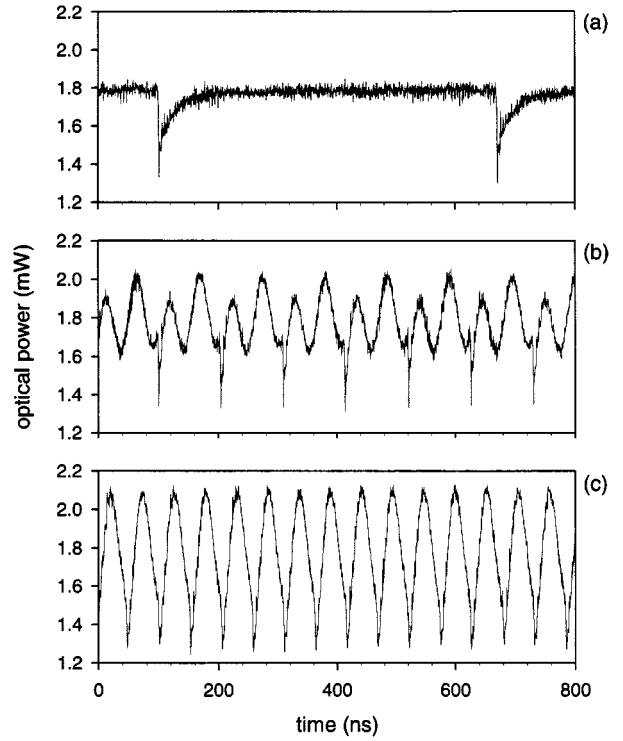


Fig. 3. Experimental time series of the laser intensity, illustrating entrainment of power dropouts. (a), (b), and (c) correspond to the same experimental conditions as Fig. 2(a), (e), and (f), respectively.

We remark that the probability distribution shown in Fig. 2(b) is also rather similar to that observed by Bulsara *et al.* [53], who investigated theoretically an integrate-fire model of a neuron driven simultaneously by periodic and noisy signals. By determining the first passage time probability distribution for the time between neuronal firings, they found that the ability of the neuron to respond to the weak periodic signal was enhanced by the presence of the noisy signal under conditions of stochastic resonance. Their model, with only stochastic driving, is similar to the model developed by Henry and Kazarinov [9] for describing the statistics of the LFF power-dropout events, suggesting a connection between our observation and stochastic resonance. Taking this analogy further, it may be possible that an ECSL operating in the LFF regime may find applications as a sensitive detector of weak periodic signals.

In our experiments, we also measure the effects on the LFF statistics as the modulation strength increases [Fig. 2(c)–(f)]. We find that the dropouts occur at increasingly shorter intervals (note the changing scales), but still occur preferentially at multiples of the drive period. Fig. 2(e) and (f) show particularly interesting cases for which $\eta(\tau)$ is nonzero only in a narrow region near a single dominant interval τ . In Fig. 2(e), this interval is 105 ns, twice the modulation period, and in Fig. 2(f) it is 53 ns. In these cases, the single-spike distribution indicates that all τ are the same, i.e., the dropouts occur at regular intervals. This is qualitatively different from the probabilistic behavior of the unperturbed system.

The time domain behavior in the case of entrainment is illustrated in Fig. 3. Fig. 3(a) is a representative sample taken from the unperturbed system, showing the usual dropout and recovery

dynamics under the same conditions as Fig. 2(a). Fig. 3(b) corresponds to the conditions of Fig. 2(e) (8% modulation amplitude) and shows that the power-dropout cycle is entrained such that one event occurs every two drive cycles. Each dropout occurs at the same phase of the drive cycle, giving rise to the spike in $\eta(\tau)$ at 105 ns seen in Fig. 2(e). Finally, Fig. 3(c) shows the case for which one dropout occurs every drive cycle, corresponding to the conditions of Fig. 2(f) (13.8% modulation amplitude).

Additional information can be obtained by examining the RF power spectra of the detected laser intensity. We find that the low-frequency noise due to irregular dropouts can be reduced by up to 20 dB when the dropouts are entrained. However, we find that the characteristic RF peaks at the external-cavity fundamental (212 MHz for this system) and higher harmonics persist in all cases, indicating that the fast pulsing dynamics [5]–[7] are not eliminated by the slow modulation.

IV. THEORETICAL ANALYSIS: LOW-FREQUENCY MODULATION

One interesting aspect of our experimental observations is that the effect of the injection current modulation on the dropouts is most pronounced when the modulation frequency is in the range of 20 MHz. This result is somewhat surprising since the typical characteristic frequencies of the system are considerably higher. For example, the relaxation oscillation frequency of the laser in the absence of external feedback is of the order of 2 GHz, and the free spectral range of the external cavity is approximately 200 MHz.

As shown in Fig. 2(c), strong modulation in the vicinity of 20 MHz forces the occurrence of dropout events. Since the antinodes of the ECSL have been implicated in the initiation of a dropout [15], we suspect that the modulation enhances the coupling between the ECSL modes and antinodes, thereby destabilizing the system. While a complete mathematical proof of this conjecture is beyond the scope of the present paper, we demonstrate that the system does indeed possess characteristic frequencies of the order of 20 MHz through an analysis of the single-mode Lang–Kobayashi rate equations [54] describing the dynamics of a semiconductor laser with weak external feedback. We first introduce the model, then perform an analysis of the steady-state solutions of the model in the absence of injection current modulation, and follow up with numerical simulations of the laser with modulation.

The single-mode Lang–Kobayashi rate equations are given by [54]

$$\frac{d\mathcal{E}}{dt} = \frac{1}{2} (1 + i\alpha) G_{\mathcal{N}}(\mathcal{N} - \mathcal{N}_{\text{th}}) \mathcal{E} + \frac{\gamma}{\tau_{\text{in}}} e^{-i\omega_o \tau_c} \mathcal{E}(t - \tau_c) \quad (1)$$

$$\frac{d\mathcal{N}}{dt} = J - \frac{\mathcal{N}}{\tau_s} - \left[\frac{1}{\tau_p} + G_{\mathcal{N}}(\mathcal{N} - \mathcal{N}_{\text{th}}) \right] |\mathcal{E}|^2 \quad (2)$$

where $\mathcal{E}$ is the slowly varying complex electric field envelope and $\mathcal{N}$ is the carrier density. The solitary laser is assumed to oscillate in a single longitudinal mode with angular frequency ω_o and to have a threshold carrier density of $\mathcal{N}_{\text{th}}$. In these equations, τ_p , τ_s , τ_{in} , and τ_c are the photon lifetime, the carrier lifetime, the round-trip time in the solitary laser cavity, and the

round-trip time in the external cavity, respectively. The feedback is characterized by γ , where γ^2 is the power reflected from the external cavity relative to that reflected from the laser mirror. The term α is the linewidth enhancement factor, and J is the current density injection rate, proportional to the injection current I . The differential gain $G_{\mathcal{N}}$ is a constant defined as $G_{\mathcal{N}} = (\partial G / \partial \mathcal{N})_{\text{th}}$, where

$$G(\mathcal{N}) = G(\mathcal{N}_{\text{th}}) + G_{\mathcal{N}}(\mathcal{N} - \mathcal{N}_{\text{th}}), \\ = \frac{1}{\tau_p} + G_{\mathcal{N}}(\mathcal{N} - \mathcal{N}_{\text{th}}) \quad (3)$$

is the gain per unit time. We use parameter values representing our experimental configuration and are consistent with measured values for SDL lasers [55], [56]. Specifically, we use: $\tau_p = 4.5$ ps, $\tau_s = 700$ ps, $\tau_{\text{in}} = 3.9$ ps, $\tau_c = 4.7$ ns, $\alpha = 4$, and $G_{\mathcal{N}} = 2.6 \times 10^{-6}$ cm³/s. We determine the threshold carrier density $\mathcal{N}_{\text{th}}$ by insisting that the steady-state solution to (1) and (2) in the absence of external feedback properly predict the observed scaling of the output power as a function of the injection current. An analysis of the equations with $\gamma = 0$ reveals that

$$\mathcal{N}_{\text{th}} = \frac{2\mathcal{P}\tau_s n}{\hbar\omega_o c \tau_p A(1 - R - \ell)(J/J_{\text{th}} - 1)} \quad (4)$$

where $\mathcal{P}$ is the power of the beam generated by the laser in the absence of external feedback, n is the refractive index of the structure, c is the speed of light in vacuum, A is the cross-sectional area of the optical cavity, R is the power reflection coefficient of the output facet, ℓ is the distributed losses in the active region, J_{th} is the value of the injection carrier density rate at the solitary laser threshold, and the reflectivity of the rear laser facet is assumed to be near one. For $I/I_{\text{th}} = J/J_{\text{th}} = 2$, we observe $\mathcal{P} = 16.3$ mW. Using these values and $n = 3.4$, $A = 3 \times 10^{-8}$ cm², and $R = \ell = 0.05$, we estimate that $\mathcal{N}_{\text{th}} = 8.8 \times 10^{16}$ cm⁻³. In the following analysis, it is convenient to use a dimensionless form of the Lang–Kobayashi equations that reduces the stiffness of the equations during numerical integration [57]. This formulation is given by

$$\frac{dE}{d\tilde{t}} = (1 + i\alpha)NE + \kappa e^{-i\Omega_o \theta_c} E(\tilde{t} - \theta_c) \quad (5)$$

$$T \frac{dN}{d\tilde{t}} = [P_o + P' \sin(\Omega_m \tilde{t})] - N - (1 + 2N)|E|^2 \quad (6)$$

where $E = (\tau_s G_{\mathcal{N}}/2)^{-(1/2)} \mathcal{E}$ is the dimensionless slowly varying complex electric field amplitude, $N = (\tau_p G_{\mathcal{N}}/2)(\mathcal{N} - \mathcal{N}_{\text{th}})$ is the dimensionless excess carrier number, $P_o = (\tau_p G_{\mathcal{N}} \mathcal{N}_{\text{th}}/2)(J/J_{\text{th}} - 1)$ is the excess dc injection current, and $\tilde{t} = t/\tau_p$. The other parameters are given by: $\kappa = \gamma \tau_p / \tau_{\text{in}}$, $T = \tau_s / \tau_p$, $\theta_c = \tau_c / \tau_p$, and $\Omega_o = \omega_o \tau_p$. The pump current modulation is added explicitly to the model by means of a dimensionless modulation amplitude P' and angular frequency $\Omega_m = \omega_m \tau_p (= 2\pi f_m \tau_p)$.

The usual phase-space model of LFF [15] based on the Lang–Kobayashi equations shows that the dynamics evolve along the external cavity modes of the system, which are the steady-state solutions of (5) and (6). These states can be characterized in terms of $|E|^2$, N , and $\Delta\Omega$, the dimensionless photon

number, carrier number, and optical angular frequency shift, respectively, where $\Delta\Omega = \Omega - \Omega_o$ and Ω represents the dimensionless optical angular frequency. Expressions that define sets of steady-state values ($|E_s|^2, N_s, \Delta\Omega_s$) can be obtained by assuming solutions of the form $E(\tilde{t}) = E_s \exp(i\Delta\Omega_s \tilde{t})$ and $N(\tilde{t}) = N_s$. With $P' = 0$, they are given by

$$|E_s|^2 = \frac{P_o - N_s}{1 + 2N_s} \quad (7)$$

$$N_s = -\kappa \cos(\Omega_s \theta_c) \quad (8)$$

$$\Delta\Omega_s = -\kappa[\alpha \cos(\Omega_s \theta_c) + \sin(\Omega_s \theta_c)]. \quad (9)$$

The modes and antimodes of the external cavity laser correspond to the solutions of the transcendental (9), where the antimodes correspond to the condition [10]

$$1 + \kappa \theta_c \sqrt{1 + \alpha^2} \cos(\Omega_s \theta_c + \tan^{-1} \alpha) < 0. \quad (10)$$

The steady-state carrier density (optical power) has an absolute minimum (maximum) for $\Omega_s \theta_c = 0 \pmod{2\pi}$, for which $\Delta\Omega_s = -\kappa\alpha$. A mode of the ECSL will occur precisely at the absolute minimum of N_s when $\Omega_o \theta_c = \kappa\alpha \pmod{2\pi}$, which can be obtained experimentally by adjusting the external-cavity length over a half-wavelength, or by adjusting slightly the dc injection current or temperature. This particular solution is referred to as the “maximum gain mode.” For simplicity, we assume that the cavity is adjusted to satisfy this condition.

To motivate why low frequencies might be successful in coupling the modes and antimodes, we investigate their steady-state frequencies in the absence of current modulation ($P' = 0$). In the following discussion, we use $\kappa = 0.088$ corresponding to the 17% reduction in the laser threshold observed in the experiments. By determining the solutions to (9) using numerical methods, we find that the difference in frequency between the maximum gain mode ($\Omega_s \theta_c = 0$) and closest antimode is equal to 16.7 MHz for $\theta_c = 1049$, and $P_o = 0.149$ (corresponding to $J/J_{th} = 1.29$). As $\Omega_s \theta_c$ decreases, the spacing between each mode and its corresponding antimode increases from 16.7 MHz to its maximum value of approximately 106 MHz as $\Omega_s \theta_c$ approaches $\pi/2$ (at the minimum linewidth mode). We conjecture that the periodic modulation impresses side-bands on each of the cavity mode and antimode frequencies, and that a resonance occurs when the modulation frequency is equal to the mode–antimode difference frequency. Exciting this resonance enhances the probability of inducing a dropout event. The precise optimum modulation frequency (in the range between 16.7 and 106 MHz) depends on the location in phase space where the trajectory resides the longest. Future investigations of this conjecture will require a complete analysis that explicitly takes into account the current modulation.

To determine whether the Lang–Kobayashi model with modulation can give rise to qualitatively similar behavior to that seen in the experiments, we numerically integrate (5) and (6). The temporal evolution of the power emitted by the ECSL is shown for no modulation in Fig. 4(a), and entrained dropout events in

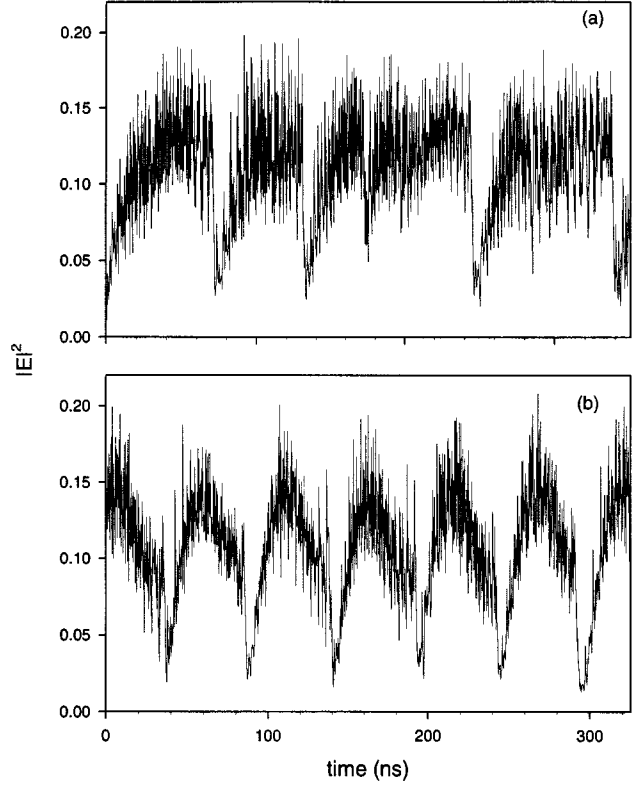


Fig. 4. Numerically generated time series of the laser intensity (a) without modulation and (b) with 6% modulation at $f_m = 19$ MHz. The periodic modulation entrains the dropout events in a fashion similar to that observed in the experiments.

Fig. 4(b) for a modulation strength of 6% ($P' = 8.9 \times 10^{-3}$) and frequency $f_m = 19$ MHz. The parameter values used in the simulation are $P_o = 0.041$ (corresponding to $J/J_{th} = 1.08$), $\theta_c = 1049$, $T = 156$, $\kappa = 0.088$, and the time series is low-pass filtered with a 250-MHz cutoff frequency. We use a lower value of the dc injection current in the simulations because the model did not display dropout events far above threshold; however, our prediction that dropout events can be entrained is rather insensitive to the precise value of the injection current. Fig. 4(a) shows a typical LFF time series that is qualitatively similar to the experimental observations shown in Fig. 3(a). Note that the dropout events occur more frequently and the intensity shows higher frequency structure in the simulations than that observed in the experiments, which we attribute to our neglect of spontaneous emission in the model [16]. For a modulation strength of 6% (Fig. 4(b)), the simulations reveal that a dropout event occurs for every modulation cycle when the laser power is near a minimum, which is similar to the experimentally observed time series shown in Fig. 3(c). Better quantitative agreement between the simulations and experiments might require the inclusion of nonlinear gain saturation effects and spontaneous emission.

Further insight concerning the laser behavior can be obtained by viewing the dynamics in phase space. The LFF cycle in the absence of modulation is shown in Fig. 5(a), projected onto the space of excess carrier number N and round-trip phase difference $\Delta\Omega\theta_c \equiv \Omega\theta_c - \Omega_o\theta_c$. It is seen that the average intensity builds up as the system drifts from the solitary laser state (upper right, $N = 0$, $\Delta\Omega = 0$) toward the maximum gain mode

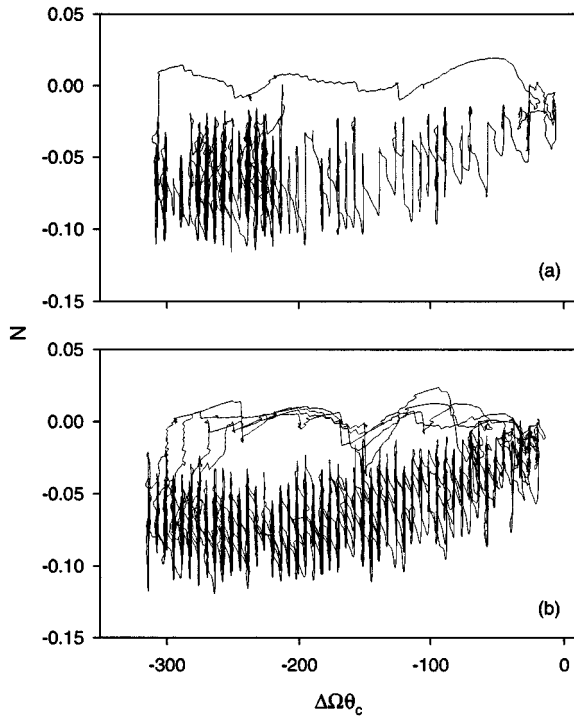


Fig. 5. Numerical phase space trajectories of the (a) unperturbed and (b) entrained dynamics of the LFF system. Graph (a) shows a single typical dropout event, and (b) shows four entrained dropouts in sequence.

(lower left, $N = -\kappa = -0.088$, $\Delta\Omega\theta_c = -\kappa\alpha\theta_c = -369.2$) during chaotic itinerancy [58]. The dropout event occurs when the trajectory comes into crisis with an unstable saddle-type antinode and is repelled back to the solitary laser state, only to repeat the process. As shown in Fig. 5(b), the modulation induces the crisis without a significant change in the extent of region in phase space visited by the trajectory. We note that the crisis occurs close to the location of the maximum gain mode where the mode–antinode frequency is nearly equal to the modulation frequency, as discussed above. Interestingly, the dynamics do not repeat *precisely* from cycle to cycle; the phase space trajectories may visit different modes and may collide with one of several nearby saddles. The overall shape and timing, however, remain quite similar.

V. EXPERIMENTAL OBSERVATIONS: HIGH-FREQUENCY MODULATION

We now consider the effects of weak open-loop perturbations on the ECSL when the frequency is in the neighborhood of the fundamental external-cavity frequency (212 MHz, as measured from the RF spectrum of the intensity fluctuations), another natural time scale present in the system. For this set of experiments, we hold the peak-to-peak amplitude of the current modulation constant at 1.9% of the dc level and vary the frequency f_m above and below 212 MHz. In this experiment, the ECSL's threshold is reduced by 15% from its solitary value by the optical feedback, and the pump level relative to threshold is set at $J/J_{th} = 1.21$. The experimentally measured probability distributions are shown in Fig. 6. As before, the first figure [Fig. 6(a)] is a reference distribution measured from the unperturbed system. Fig. 6(b)–(f) shows the results obtained as f_m is

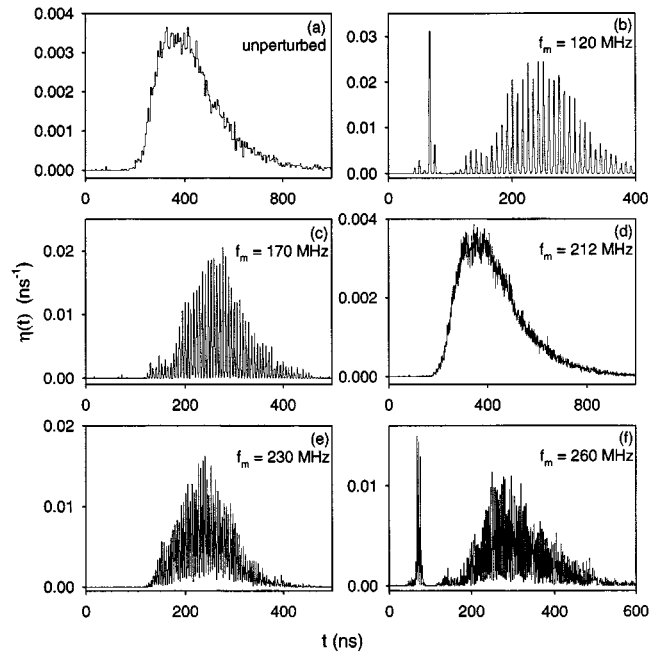


Fig. 6. Experimental probability distributions $\eta(\tau)$ of dropout intervals with high-frequency current modulation of fixed amplitude, 1.9% of dc. (a) Unperturbed distribution. Modulation frequencies are (b) 120 MHz, (c) 170 MHz, (d) 212 MHz, first cavity resonance, (e) 230 MHz, and (f) 260 MHz.

increased from 120 to 260 MHz. Three interesting effects are apparent. First, the weak modulation creates a comb in $\eta(\tau)$ in all but one case, and the spikes of the comb are separated by the period of the drive. Thus, as in the low-frequency case, the power dropouts occur preferentially at one phase in the drive cycle. In addition, the mean dropout interval $\langle\tau\rangle$ shifts to shorter times. The second effect is that $\eta(\tau)$ is unchanged in all respects when f_m is set precisely to 212 MHz, the first cavity resonance. The shape of the distribution with $f_m = 212$ MHz and when the laser is unperturbed are the same, and $\langle\tau\rangle$ is also the same within the statistical uncertainty ($\sim 1\%$). This phenomenon can be observed in numerical simulations of the Lang–Kobayashi equations as well, but its origin is not yet fully understood. We note that Takiguchi *et al.* [40] observed that dropouts can be suppressed when the modulation frequency is equal to the external-cavity mode frequency, contrary to our observations. We do not know the reason for this discrepancy, although the experimental configuration is quite different in the two situations.

The final experimental observation is that the envelope of the distribution acquires a sharp peak at short times (~ 50 ns) for $f_m = 120$ and 260 MHz. An analysis of the time series giving rise to these distributions reveals that the dropouts occur in short rapid bursts at a frequency close to 20 MHz.

VI. POSSIBLE APPLICATION TO CHAOS COMMUNICATION

Before stating our conclusions, we digress briefly to suggest that the ECSL with periodic modulation might serve as a building block for a new method of communicating information using complex dynamical systems. Recently, researchers have demonstrated that it is possible to transmit a private message by encoding it within a chaotic carrier and recovering the message using a receiver that is dynamically similar to the transmitter

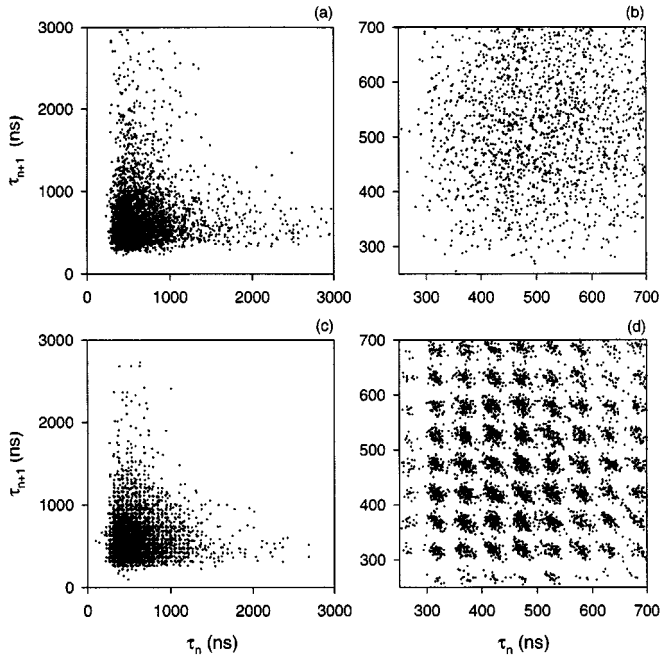


Fig. 7. Phase-space reconstruction of the experimentally observed dropout intervals obtained by plotting the n th + 1 interval as a function of the n th interval. (a) Phase space reconstruction of the data shown in Fig. 2(a) for the case of no periodic modulation. (b) High-resolution plot of (a). (c) Phase-space reconstruction of the data shown in Fig. 2(b) for the case of periodic modulation of the injection current with a peak-to-peak amplitude of 0.7% of the dc level. (d) High-resolution plot of (b), demonstrating distinct phase-space partitions.

[59]. One particular class of chaos communication schemes relies on perturbing the transmitting system such that it follows a prescribed symbol sequence; the allowed symbol sequence (called the grammar) is then accessed to encode any desired message [60]. For this method to be robust against noise, the phase space of the dynamical system must be partitioned into distinct well-separated regions [61]. The main strength of this communication scheme is that the high-power nonlinear chaotic oscillator generating the transmission signal can remain simple and efficient while the control perturbations can remain small. In addition, the scheme can withstand larger levels of communication channel noise in comparison to other methods [62]. Note that this communication scheme is not necessarily private since any observer can detect the symbols, although modifications to the method have been shown to increase its privacy [63].

We suggest that the phase space spanned by the n th interval τ_n between dropout events and the $(n+1)$ interval τ_{n+1} generated by the ECSL laser with periodic modulation provides a useful partition for communication with chaos. Fig. 7 shows such a phase-space reconstruction for the ECSL without Fig. 7(a) and with Fig. 7(c) periodic modulation, corresponding to the probability distributions shown in Fig. 2(a) and (b), respectively. Without modulation, the attractor shows no discernible structure, indicating that the dynamics are highly dimensional, which may be important for increasing the privacy of the communication scheme. With modulation, the phase space breaks up into distinct partitions, where each region would correspond to a distinct symbol.

Our suggestion is supported by the recent research of Alsing *et al.* [63], who investigate the use of interspike intervals in a

chaotic solid-state Nd:YAG laser (analogous to the interval between dropouts in the ECSL) for chaos communication. They find it is possible to encode and decipher messages using this scheme, although the time scale of the intervals in the solid state is much longer than that in the semiconductor laser studied here. To ascertain whether the ECSL device is useful for chaos communication, additional research is needed to determine the grammar of the dynamics, and whether small perturbations can reliably transfer the system from one symbol to the next on such fast time scales. Recent theoretical studies by Naumenko *et al.* [29] suggest that such control of the semiconductor laser dynamics is possible.

VII. CONCLUSIONS

We have performed an experimental and theoretical investigation of the effects of current modulation on the statistics of power dropouts in an ECSL for two different regimes of the modulation frequency. We find that weak perturbations change the probability densities of the dropout interval times over a wide range of modulation frequencies, creating a comb in the time-interval probability distributions with peaks separated by the modulation drive period. This indicates that the dropouts occur preferentially at a particular phase of the drive cycle at a given frequency, even though the total number of cycles between dropouts may vary greatly. Such an effect may be useful in small-signal detection, or in an optical chaos-based encryption scheme.

We have also demonstrated that strong modulation at a frequency in the vicinity of 20 MHz can entrain the dropouts, eliminating the otherwise probabilistic timing between events and suppressing low-frequency RF noise. This entrainment can occur such that one dropout occurs each drive period or every other drive period. We conjecture that the effect of the modulation on the dropouts is most pronounced when the modulation frequency is equal to the frequency difference between the ECSL's mode and its adjacent antinode. High-frequency modulation can also partially entrain the dropout but requires large modulation amplitudes. Finally, we find that there is no effect on the dropout interval distribution when the modulation frequency is exactly equal to the external-cavity round-trip frequency. We hope that future research may answer some of the questions raised by these experiments and that those answers may give insight into the mechanisms of LFF. Specifically, it may be instructive to study the mechanism by which the dropouts select a preferred drive cycle phase and to understand why modulation at the first cavity resonance leaves the dropout intervals unaffected.

ACKNOWLEDGMENT

The authors gratefully acknowledge discussions of the SDL laser parameters with T. Simpson.

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